

Superior mechanical performance polyurethane-silicone coatings for synergistic antimicrobial and antifouling coatings

Ming Liu, Laiying Jing, Ziqi Zhang, Xin Li, Huaming Yuan, Jiaxi Cui, Mei Li & Yunqiang Zhang

To cite this article: Ming Liu, Laiying Jing, Ziqi Zhang, Xin Li, Huaming Yuan, Jiaxi Cui, Mei Li & Yunqiang Zhang (2024) Superior mechanical performance polyurethane-silicone coatings for synergistic antimicrobial and antifouling coatings, Journal of Macromolecular Science, Part A, 61:4, 241-251, DOI: [10.1080/10601325.2024.2330600](https://doi.org/10.1080/10601325.2024.2330600)

To link to this article: <https://doi.org/10.1080/10601325.2024.2330600>

 View supplementary material 

 Published online: 26 Mar 2024.

 Submit your article to this journal 

 Article views: 283

 View related articles 

 View Crossmark data 

 Citing articles: 5 View citing articles 



Superior mechanical performance polyurethane-silicone coatings for synergistic antimicrobial and antifouling coatings

Ming Liu^a, Laiying Jing^a, Ziqi Zhang^a, Xin Li^a, Huaming Yuan^a, Jiaxi Cui^b, Mei Li^a, and Yunqiang Zhang^a

^aSchool of Materials Science and Engineering, Qilu University of Technology (Shandong Academy of Sciences), Jinan, P.R. China; ^bInstitute of Fundamental and Frontier Sciences, University of Electronic Science and Technology of China, Chengdu, P.R. China

ABSTRACT

With the increasing awareness of marine ecological environment, biocide-free marine antifouling coatings have been gaining increased attention in the field of marine antifouling. Among all the antifouling approaches, constructing fouling-release coatings (FRCs) may be the most effective and economical way caused by the biocide-free. However, the low mechanical strength, weak adhesion to substrates, and poor static fouling resistance of traditional silicone-based FRCs, which limiting their applications. In this work, we synthesized a polyurethane with hydroxyl terminated polydimethylsiloxane (PDMS), polytetrahydrofuran (PTMG) and isophorone diisocyanate (IPDI) as main chains, 2,2'-Bis(hydroxymethyl)butyric acid (DMBA) as chain extender, and γ -aminopropyl triethoxysilane (KH-550) as crosslinker by condensation polymerization. Meanwhile, Triclosan (TCS) was used as an antibacterial agent and mixed into the polymer system with uniformly dispersed. We introduced urethane groups into silicone, which improved the mechanical strength (ultimate tensile strength: 1.379 MPa, elongation at break: 282%). As a result, the laboratory bioassays using bacteria (*E. coli*, and *S. aureus*) showed that the anti-bacterial efficiency could respectively reach $\sim 95\%$ and $\sim 93\%$, which demonstrated an outstanding antifouling property of the D:K = 0.7:1.3/1.0. This work overcomes the low strength issue of traditional fouling-release coatings and provides a simple strategy for silicone-based polyurethane antifouling coatings.

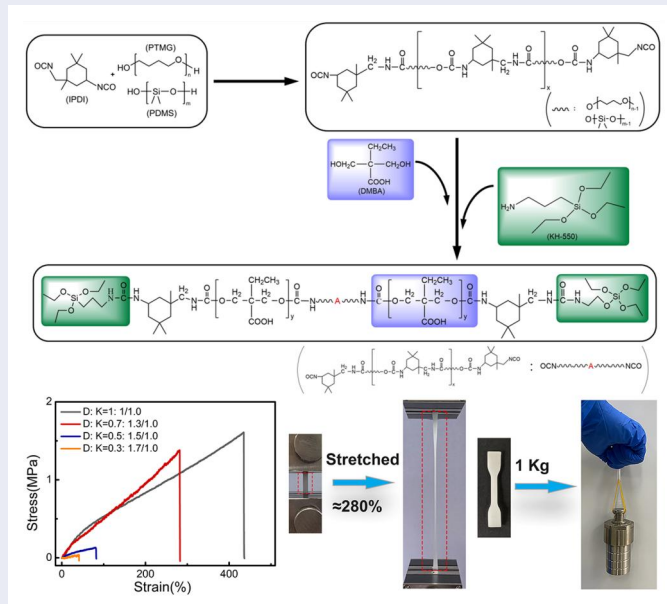
ARTICLE HISTORY

Received January 2024
Accepted March 2024

KEYWORDS

Polydimethylsiloxane (PDMS); polyurethane; marine antifouling; low surface energy; antifouling coatings

GRAPHICAL ABSTRACT



1. Introduction

Marine fouling refers to the biological pollutant formed by the attachment, growth, and reproduction of marine microorganisms, animals, and plants on the surface below seawater, which brings many problems to marine facilities and marine development.^[1–3] Biological pollutant enhances the ship's navigation resistance, reduces its speed, and increases fuel consumption.^[4] While the attached organisms may cause biological invasion problems as ships migrate to other seas.^[5,6] In addition, the biological pollutant can also accelerate the corrosion of metal surfaces,^[7] bringing safety hazards to offshore mining equipment such as offshore oil production platforms and blocking the mesh holes of the cages for marine farming, hinder the delivery of nutrients and reduce production.^[8–11] Therefore, it is crucial to prevent and control marine fouling during marine development.

Among the many marine antifouling methods, antifouling coatings are widely applied because of their cost-effectiveness and convenience.^[12] Antifouling coatings can be divided into biocidal and non-biocidal types in accordance with the release of biocides.^[13] The biocidal antifouling coatings release biocides to disperse or kill fouling organisms.^[14,15] However, the released biocides has caused significant harm to the marine ecosystem.^[16,17] With the awakening of environmental awareness, biocides have been strictly reviewed in many countries,^[18,19] and non-biocidal antifouling coatings have received wide attention, among which the most promising is fouling-release coatings (FRCs).^[20] Silicone elastomers are widely used in FRCs, especially polydimethylsiloxane (PDMS), since they have low surface energy (SE) and surface roughness, which makes it difficult for dirt and biological matter to adhere to the coated surface, and they are easily detached under fluid dynamic stress or mechanical cleaning.^[21–23] However, PDMS have poor mechanical properties,^[24] substrate adhesion,^[25] and poor antifouling performance under static conditions which significantly limits its applications.^[24–26]

To circumvent the above issues, some effective solutions were proposed to improve the mechanical strength, adhesion to a substrate, and fouling resistance, such as introducing epoxy or urethane/urea groups, amphiphiles, zwitterions, quaternary ammonium salts (QASs), and antifoulants.^[13,27,28] Peng et al. reported a biologically inspired self-layered coating, which consists of polyurethane with dopamine and silyl groups, poly(dimethylsiloxane) terminated with fluorosilicon compounds and polyethylene glycol segments and polyethylene glycol segments as a reactive amphiphilic polymer.^[29] The coating exhibits strong substrate adhesion and excellent corrosion resistance because of the dopamine and silane groups. However, the complex synthesis process limits its further application. Sheng et al. developed a scratch-resistant self-healing waterborne polyurethane-urea coating with a dual metal coordination structure of Zn(II) carboxylate and Zn(II) diaminopyridine.^[30] The incorporation of zinc ions enables the formation of a dense metal coordination bond on the coating's surface, significantly enhancing the coating's scratch resistance and surface hardness (from B to 4H).

Nevertheless, the existence of metal ions may cause environmental pollution.^[31] In this regard, it is still a challenge to design a FRC that simultaneously has high mechanical strength and excellent static antifouling ability.

In the present study, PDMS-based polyurethane elastomers were fabricated by a facile method using polydimethylsiloxane (PDMS), polytetrahydrofuran (PTMG), 2,2'-Bis(hydroxymethyl) butyric acid (DMBA), γ -aminopropyl triethoxysilane (KH-550) and triclosan (TCS). Since PDMS segments are covalently bonded with DMBA, the coating exhibits not only good flexibility but also outstanding elongation and high mechanical strength. In addition, the presence of TCS could significantly improve the antibacterial performance of the elastomer under static conditions, which act on both Gram negative and Gram positive bacteria.^[32] Owing to the superior mechanical properties (The best comprehensive performance sample's ultimate tensile strength: 1.379 MPa, elongation at break: 282%) and antifouling performance (The anti-bacterial efficiency of *E. coli*, and *S. aureus* could respectively reach ~95% and ~93%), this elastomer has broad prospects for applications in the field of marine antifouling coatings.

2. Experimental

2.1. Materials

Sianol terminated Polydimethylsiloxane, 45-85 cSt (PDMS) was purchased by Gelest. Isophorone Diisocyanate (IPDI), Polyteramethylene glycol (PTMG, Mn = 2000 g/mol), Dibutyltin dilaurate (DBTDL), γ -aminopropyl triethoxysilane (KH-550) were supplied by Aladdin. 2,2'-Bis(hydroxymethyl) butyric acid (DMBA) and Triclosan (TCS) were purchased from Macklin. Acetone was purchased from Yantai Yuan Dong Fine chemicals Co., Ltd.

2.2. Fabrication of PDMS-based polyurethane

The PDMS-based polyurethane was synthesized *via* condensation polymerization, in which DMBA and KH-550 act as chain extender and crosslinker, respectively (Figure 1). PTMG (3.7500 mmol, 7.5000 g) was dissolved in acetone, and pour the solution into a three-necked round-bottom flask equipped with a condenser and a stirrer. Then, PDMS (3.7500 mmol, 10.3125 g) were added into the flask. After stirring well, IPDI (9.7500 mmol, 2.1700 g) and DBTDL (0.0500 g) were added to the flask and then stirred at 80 °C for 2 h. DMBA (0.9545 mmol, 0.1414 g) was added into the flask and then stirred at 80 °C for 2 h. Finally, KH-550 (1.7727 mmol, 0.3924 g) was added into the flask and then stirred at 80 °C for 2 h. PDMS-based polyurethane with varying TCS contents were prepared using a solution mixing method. For convenience, the PDMS-based polyurethane with 1.0 wt% TCS is designed as D: K = x: y/1.0, where x: y is the ratio of the amount of substance in DMBA and KH-550. The characterization data are summarized in Table S1.

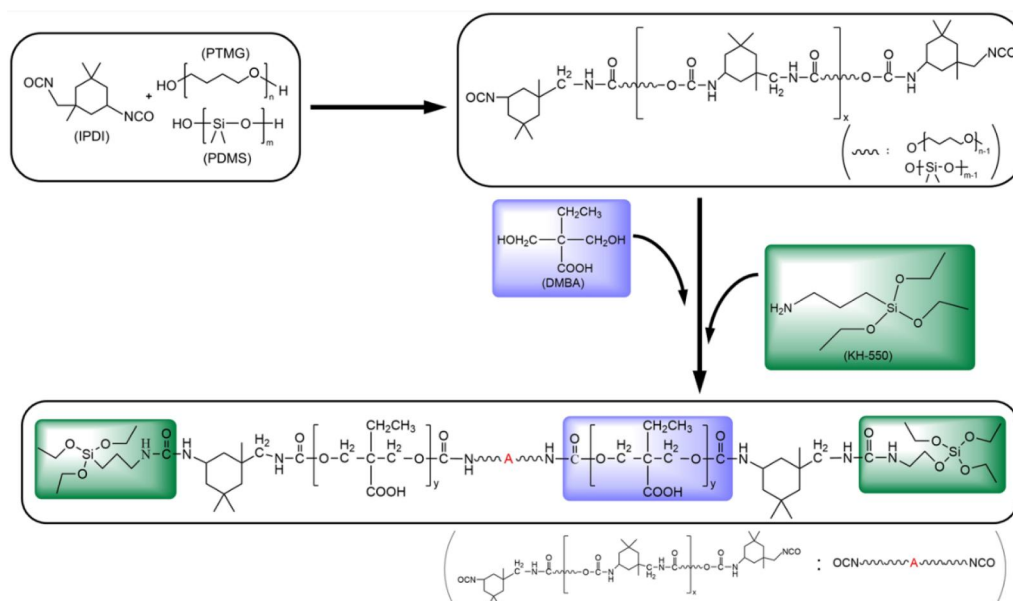


Figure 1. Synthesis of PDMS-based polyurethane.

2.3. Characterization

Attenuated Total Refraction (ATR) was detected using a Nicolet 10 spectrometer under room temperature conditions. The sample was scanned and analyzed 32 times within the wavelength range of 4000 cm^{-1} to 400 cm^{-1} . Nuclear Magnetic Resonance Hydrogen Spectrum (^1H NMR) was detected using JNM-ECZ400S at a frequency of 400 MHz under room temperature conditions. All samples were dissolved in CDCl_3 . For thermogravimetric analysis (TGA), TGA-1 Mettler Toledo thermogravimetric analyzer was employed. The sample were conducted under nitrogen gas atmosphere. The heating rate was controlled at $10^\circ\text{C}/\text{min}$ and the temperature range extended from 30°C to 800°C . The sample mass used for testing was 10 mg. The composition of elements and chemical bonds in the sample was detected using X-ray Photoelectron Spectroscopy (XPS). Energy Dispersive Spectrometer (EDS) were performed on ZEISS Gemini 500 with 15.0 kV acceleration voltages.

2.4. Contact angle and surface energy

Deionized water and n-hexadecane contact angle (CA) were measured through the German device (Kruss DSA25). The tests were performed at room temperature in $2\text{ }\mu\text{L}$ of deionized water and n-hexadecane with the static sessile drop method. The three CA values (θ) were averaged at different areas for each sample. The surface free energy (SE) was calculated by the Owens two-liquid method based on the WCA of three measurements and n-hexadecane CA of three measurements.^[33] The water and hexadecane contact-angle data for the D: K = 1: 1/1.0, D: K = 0.7: 1.3/1.0, D: K = 0.5: 1.5/1.0, D: K = 0.3: 1.7/1.0 were measured, and the results are shown in Table 1.

Table 1. Static contact angles and surface energy of PDMS-based polyurethane coating.

Samples	$\theta_{\text{water}} (^\circ)$	$\theta_{\text{hexadecane}} (^\circ)$	γ_s (mN/m)	γ_s^d (mN/m)	γ_s^p (mN/m)
D: K = 1: 1/1.0	108.9 ± 1.1	41.4 ± 1.2	21.3	21.10	0.20
D: K = 0.7: 1.3/1.0	111.0 ± 1.0	41.2 ± 1.1	21.26	21.19	0.07
D: K = 0.5: 1.5/1.0	112.0 ± 1.1	46.4 ± 0.8	19.78	19.70	0.08
D: K = 0.3: 1.7/1.0	118.5 ± 1.3	45.6 ± 1.0	19.99	19.93	0.06

2.5. Adhesion test

The adhesion strength of the samples to substrates (steel, glass, and epoxy) was measured by using a XH-M type pull-off adhesion tester (Beijing Tiandi Spark Technology Development Co., Ltd) according to the GB/T5210-1985. An aluminum ingot was bonded to the surface of the coatings using acrylic adhesive (Cyanoacrylate adhesive, Ergo). For each sample, three values from different test points were averaged to determine the adhesion strength.

2.6. Tensile test

The stress-strain behavior of the coatings was determined using a universal tensile testing machine. The samples were cut into a standard dumbbell shaped with dimensions of $(50 \times 12.5 \times 4\text{ mm})$. The tensile rate was set at $50\text{ mm}/\text{min}$.

2.7. Shore hardness test

Using the LAC-YJ Shore A hardness tester from Shanghai Qinyun Technology Co., Ltd, according to the national standard GB2411-78-1980, the average value is taken after 10 measurements.

2.8. Swelling ratio

By weighing the initial mass of a square sample measuring $20 \times 20\text{ mm}$ as m_0 , immersing the sample in artificial

seawater 24 h, removing surface artificial seawater droplets of the sample surface and weight it as m_t . The swelling ratio was calculated by following formula:^[34]

$$(m_t - m_0)/m_0 \times 100\% \quad (1)$$

2.9. Self-cleaning test

Hydrophilic contaminants ($\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$) and hydrophobic contaminants (carbon particles) are scattered on the coated surface, and after 1 h, the contaminants are removed with water droplets.^[35,36]

2.10. Antibacterial assays

The plate count method was used to study the antibacterial performance of the coatings against *E. coli* and *S. aureus*. The coated cover slips were soaked in sterile phosphoric acid buffer solution for 24 h in advance to fully leach the possible small molecules and then sterilized with UV lamp. The coating samples on coverslips were immersed in 5 mL of *E. coli* suspension or *S. aureus* suspension (10^8 CFU/mL) and incubated in a shaking incubator at 37°C for 5 h. Gently rinse the surface three times with 10 mL of sterile phosphate buffer solution to remove weakly adhering bacteria. Take 200 μL of the final washing solution, disperse it in a solid culture medium, and incubate at 37°C for 24 h. Observe the formation of bacterial colonies and determine the antibacterial performance of the coating.

N_g is N_0 is defined as the number of bacteria on the glass slide without coating, N_s is defined as the number of bacteria on the glass slide with coating. The antibacterial rates of the coatings were calculated according to the formula:^[30]

$$(N_g - N_s)/N_g \times 100\% \quad (2)$$

2.11. Diatom settlement

Fluorescence microscopy was used to observe the growth and attachment of algae on the surface of the coating, the algae resistance of the coating was evaluated by *Chlorella*. The coverslips and coverslips coated with D: K=0.7: 1.3, D: K=0.7: 1.3/1.0 were placed in a six-well plate with 10 mL *Chlorella* suspension, and incubated in the mode of light for 12 h and dark for 12 h at 25°C . After 72 h, the unattached *Chlorella* on the surface of the coating was gently washed off by artificial seawater, the scale of the measurement was 100 μm , and the number of attached *Chlorella* was counted by image j software.

2.12. Salt spray test

The coating shall be tested in accordance with the GB/T1771-2007 standard. Apply the coating to the middle part of carbon steel, as a comparison both sides were un-coated. Add 3.5 wt% NaCl solution to a salt spray machine at 37°C , and record the degree of rust on the steel after 1 d, 3 d, 5 d, and 7 d.

3. Results and discussion

3.1. Chemical characterization of the PDMS-based polyurethane coatings

FTIR spectra were recorded in Figure 2a and b for PDMS-PU coatings. The strong absorption peak at 2950 cm^{-1} and 2854 cm^{-1} were related to the stretching vibration of the asymmetric and symmetric methyl groups.^[34] Furthermore, The disappearance of the NCO characteristic peak at 2270 cm^{-1} and the appearance of the urethane absorption peak at 1716 cm^{-1} indicate the successful synthesis of polyurethane.^[37] The peak at 1255 cm^{-1} represents the in-plane bending vibration peak of Si-CH₃. The peaks at 1000 and 785 cm^{-1} are related to the symmetric stretching vibration of Si-O in Si-O-Si groups and the stretching vibration peak of Si-C, respectively.^[14] Moreover, the peak at 1100 cm^{-1} can be attributed to the asymmetric stretching mode of Si-O-Si.^[38] ¹H NMR spectrum of D: K=0.7: 1.3/1.0 was exhibited in Figure 2c. The methyl in PDMS (0.04 ppm), the methyl in DMBA (0.92 ppm), the methylene in DMBA (1.04 ppm), the methylene in PTMG (1.80 ppm, 3.40 ppm). The peaks at 0.04, 0.92, 1.04, 1.80, and 3.40 ppm, which belong to PDMS, DMBA, DMBA, PTMG, and PTMG, indicating PTMG and DMBA have been successfully introduced into the polymers. The thermal stability of D: K=0.7: 1.3/1.0 was studied by TGA shown in Figure 2d. From the TGA curve we can see that the main weight loss occurs above 300°C and complete decomposition occurs above 600°C . In addition, the DTG curve shows two main decomposition temperatures of the coating. The first decomposition stage between 300 and 360°C is corresponds to the pyrolysis of the hard segment and the second decomposition stage from 360 to 600°C is due to the pyrolysis of the soft segment, which also consistent with the FTIR results.^[39]

The elemental distribution of the as-obtained elastomer was shown in Figure 3. The C, N, O, Si, and ClEDS elemental mapping shows that the four elements are evenly distributed on the surface of the PDMS-PU coatings.

The survey spectrum of Figure 4a exhibits the hybrids composed of C, N, O, Si, and Cl, which agree with the EDS testing. There are three peaks at 284.42, 284.83, and 285.25 eV in Figure 4b, which can be assigned to C-C, C-O and C=O bonds, respectively.^[40] The N 1s spectrum in Figure 4c displays one peak at about 399.75 eV corresponding to C-N bonds. The O 1s spectrum (Figure 4d) shows three peaks at 531.91, 532.35 and 532.89 eV which belong to C-O, Si-O and C=O respectively. The Si2p spectrum in Figure 4e displays two peaks at about 102.25 and 102.67 eV corresponding to Si-C and Si-O, respectively. The Cl2p spectrum (Figure 4f) could be divided into two binding energy peaks at 197.58 and 200.39 eV , which are assigned to Cl 2p_{3/2} and Cl 2p_{1/2}.

3.2. Contact angle and surface energy

The wettability mechanism of the coatings was evaluated using WCA and SE. Figure 5a showed that the WCA for D: K=1: 1/1.0, D: K=0.7: 1.3/1.0, D: K=0.5: 1.5/1.0, D:

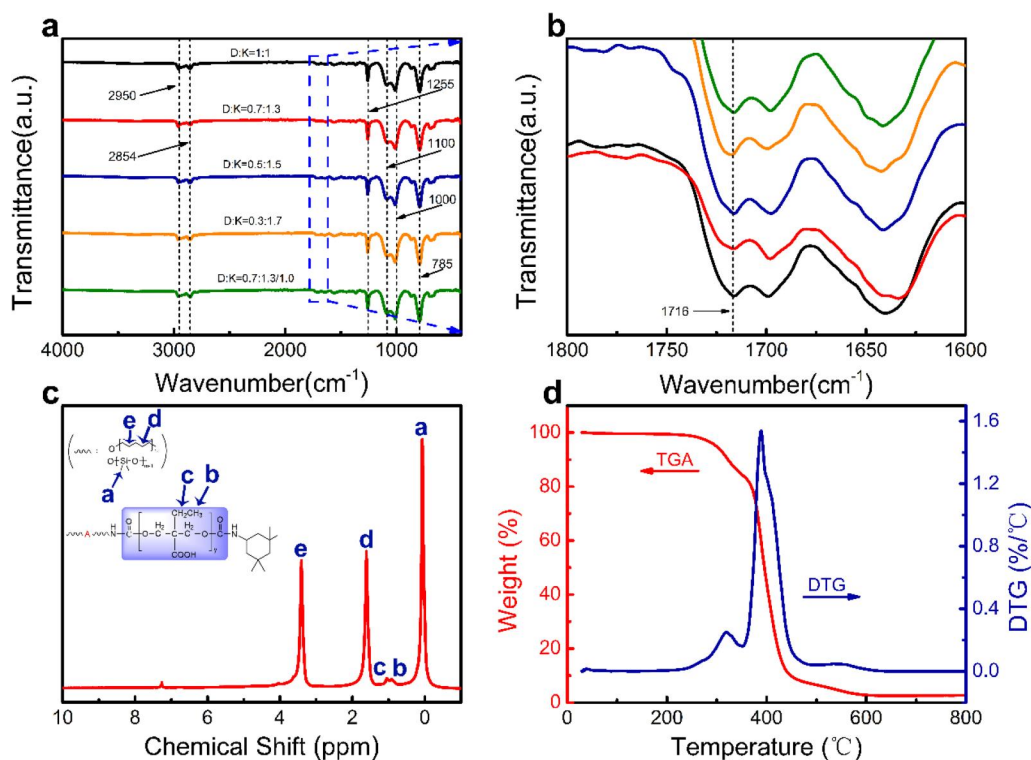


Figure 2. The chemical structure characterization of the coatings. (a) FT-IR spectra. (b) Magnified regions of (a) in the range of 1600–1800 cm⁻¹. (c) ¹H NMR of D: K = 0.7: 1.3/1.0. (d) TGA and derivative thermogravimetry (DTG) curves of D: K = 0.7: 1.3/1.0.

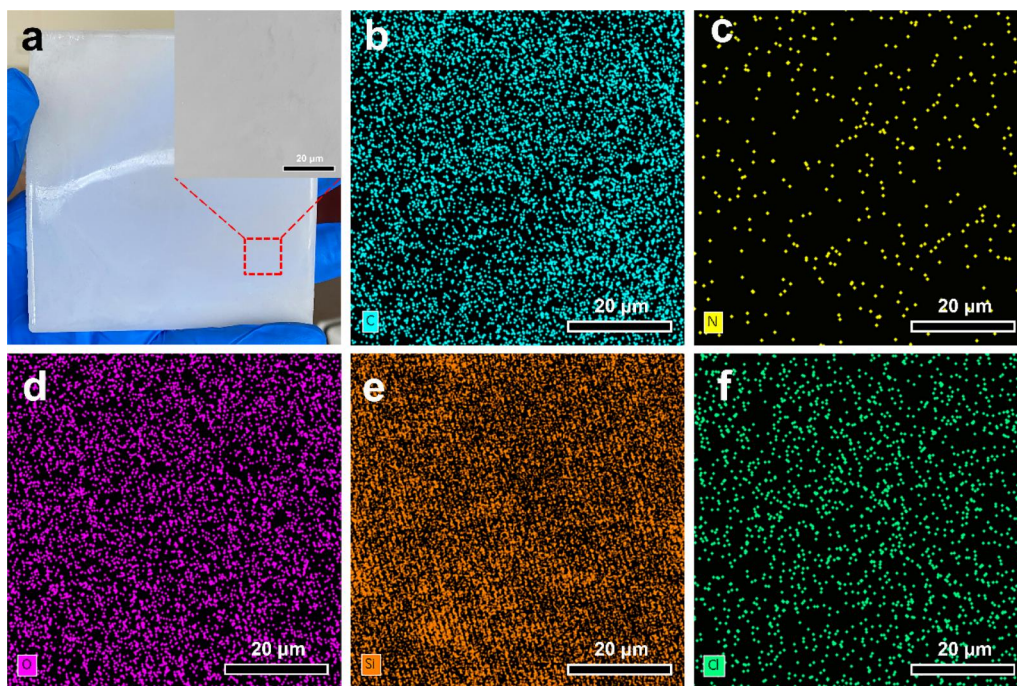


Figure 3. EDS elemental mapping images of the D: K = 0.7: 1.3/1.0. (a) SEM image. (b)–(f) the corresponding elemental mapping images of C, N, O, Si, Cl.

K = 0.3:1.7/1.0 were $108.9^{\circ} \pm 1.1^{\circ}$, $111.0^{\circ} \pm 1.0^{\circ}$, $112.0^{\circ} \pm 1.1^{\circ}$ and $118.5^{\circ} \pm 1.3^{\circ}$ respectively. The WCA gradually increases with the decreasing of the percentage of DMBA, which should be ascribed to the hydrophilic groups (-COOH) of DMBA. As for D: K = 0.7: 1.3 (Figures 2) and D: K = 0.7: 1.3/1.0 (Figure 5a), The content of TCS also exhibits a positive correlation with the WCA, which can be attributed to the hydrophobicity of TCS.

The values of the surface energy of the coating are illustrated in Figure 5b. All the samples display the surface energy range from 19.99 to 21.30 mN/m, due to the intrinsic low surface energy of PDMS segment.^[41,42] The low surface energy characteristics of PDMS-based polyurethane coatings align with a range (20–30 mN/m) conducive to the removal of settling organisms.^[21,43] This range is based on the relationship between surface energy and the fouling release

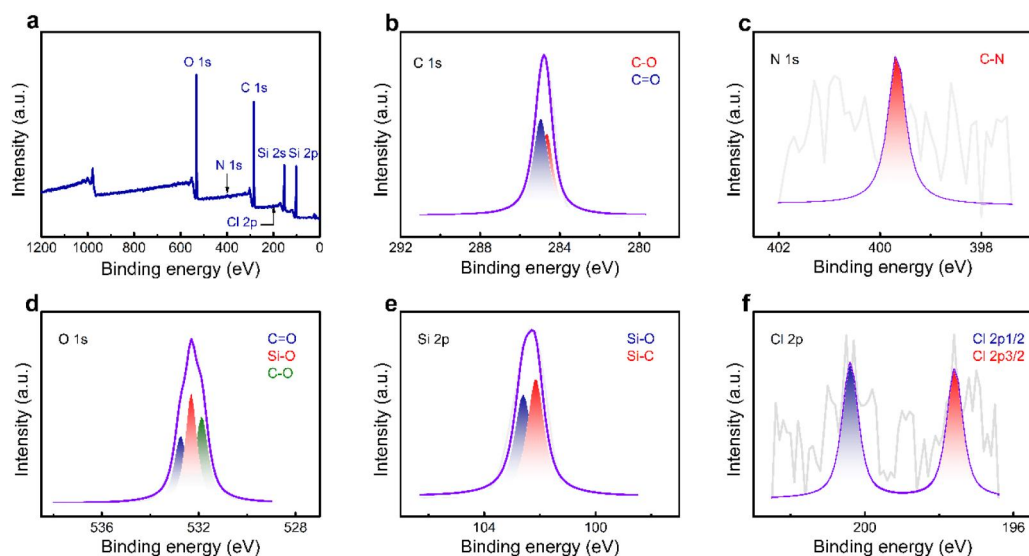


Figure 4. (a) Survey scan spectrum of the D: K = 0.7: 1.3/1.0. (b) C 1s spectra of the D: K = 0.7: 1.3/1.0. (c) N 1s spectra of the D: K = 0.7: 1.3/1.0. (d) O 1s spectra of the D: K = 0.7: 1.3/1.0. (e) Si 2p spectra of the D: K = 0.7: 1.3/1.0. (f) Cl 2p spectra of the D: K = 0.7: 1.3/1.0.

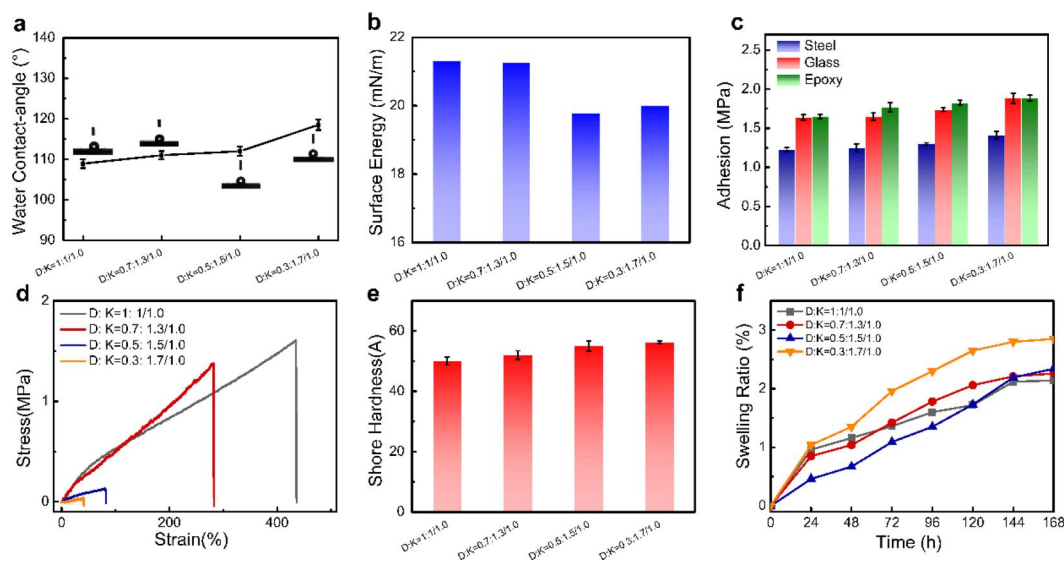


Figure 5. The mechanical properties, surface performances, adhesion strength, and swelling ratio of the coatings. (a) Water contact-angle (WCA). (b) Surface energy (SE). (c) Adhesion strength. (d) The stress and strain curves. (e) Shore Hardness. (f) Swelling ratio.

Table 2. Adhesion strength of samples.

Samples	Adhesion strength-Steel (MPa)	Adhesion strength-Glass (MPa)	Adhesion strength-Epoxy (MPa)
D: K = 1: 1/1.0	1.229 ± 0.024	1.638 ± 0.036	1.651 ± 0.028
D: K = 0.7: 1.3/1.0	1.248 ± 0.052	1.646 ± 0.048	1.769 ± 0.058
D: K = 0.5: 1.5/1.0	1.299 ± 0.016	1.736 ± 0.023	1.824 ± 0.032
D: K = 0.3: 1.7/1.0	1.409 ± 0.048	1.880 ± 0.065	1.886 ± 0.036

curve, indicating that the superior hydrophobicity of the coating.

3.3. Adhesion test

The adhesion strength of the coatings adhered to the steel, glass, and epoxy are shown in Figure 5c and Table 2. The adhesion strength to glass slightly increases with the addition of KH-550, and D:K = 0.3:1.7/1.0 exhibits the highest

adhesion strength to glass of ~1.88 MPa. Similarly, D:K = 0.3:1.7/1.0 exhibits the highest adhesion strength to steel of ~1.41 MPa and the highest adhesion strength to epoxy of ~1.89 MPa. Due to the cross-linking effect of KH-550, the hydrogen bonding between molecular chains is enhanced, so the adhesive strength to steel, glass, and epoxy increases. Comparing Figures 5c and S3, it can be obtained that as the load of TCS increases, the adhesion to glass, steel, and epoxy decreases, but the decrease is small.

3.4. Mechanical properties

Figure 5d shows the stress-strain curves of the coatings. As the addition of DMBA, the ultimate tensile strength and the elongation at break gradually increase. D:K = 1:1 has high tensile strength of 4.173 MPa, and its elongation at break is up to 853.4% (Figure S4). This is because the addition of chain extender DMBA prolongs the length of the molecular

chain and increases the ultimate tensile strength accordingly. As the proportion of crosslinking agent KH-550 increases, the molecular chains are crosslinked and solidified, limiting their fluidity, which leads to a decrease in elongation at break. Besides, with the increase in TCS load, both ultimate tensile strength and elongation at break decline seriously. The D:K=0.7: 1.3/1.0's tensile strength is up to 1.382 MPa, and its elongation at break is up to 282.6%.

The hardness of the coating is evaluated by Shore A. As a coating, good hardness is a means of protecting oneself from irreversible damage. Carbamate as a hard segment can improve the hardness of elastomers. As shown in Figure 5e, as the content of KH-550 increases, more cross-linking points are formed, and the hardness shows a continuously increasing trend. The maximum Shore hardness can reach 56.25 A (D: K=0.3: 1.7/1.0). Furthermore, the introduction of TCS reduces the hardness of the coating, and the larger the amount of introduction, the greater the decrease. The main reason that mechanical properties decrease should be the destruction of cross-linking structure by the dispersed TCS particles (Figures 5e and S5). Meanwhile, the hardness result of the coating corresponds one-to-one to the elongation at break, that is, the smaller the elongation at break, the greater the hardness.

3.5. Swelling ratio

Figure 5f presents the swelling ratio of the coatings after being immersed in artificial seawater for 7 days at room temperature. The swelling rate of all of the coatings is measured every other day. Figure 5f shows that the swelling ratio of D: K=1: 1/1.0, D: K=0.7: 1.3/1.0, D: K=0.5: 1.5/1.0 and D: K=0.3: 1.7/1.0 were 2.14%, 2.26%, 2.34% and 2.85% after reaching the swelling equilibrium. The result suggested that the change in the ratio of DMBA to KH-550 had little effect on the swelling ratio. Besides, the swelling ratio of D: K=1: 1, D: K=0.7: 1.3, D: K=0.5: 1.5 and D: K=0.3: 1.7 are given in Figure S6, it seemed to decrease to 1.52%,

1.68%, 1.60% and 2.65%, respectively, which might be due to the compactness of the coating: the introduction of TCS may have disrupted the compactness of the coating, although TCS is hydrophobic. All the coatings exhibit small swelling rate, which can be attributed to the hydrophobicity of the coating.

3.6. Self-cleaning properties

When coatings are exposed to the marine environment, abiotic contaminants in seawater also come into contact and adhere. Therefore, self-cleaning tests were conducted on pollutants with different hydrophilicity to evaluate their ability to resist the attachment of inorganic particles. $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$ is used as hydrophilic particle, and carbon particles are used as hydrophobic contaminant. Contaminants are sprinkled on coated slides and deposited for 1h, slides are placed at an inclination angle of 10° , and surface contaminants are removed using water droplets. As shown in Figure 6a–d, only a small amount of water is required to dissolve the copper chloride particles and slide down with the water flow, whereas in the case of carbon particles, apart from extremely fine carbon powder residue on the coating surface, most of the contaminants were washed away by water droplets. These results also confirm the low surface energy characteristics of the coating. The results show that abiotic pollutants can be removed from the coating surface under the water flow, and the coating shows excellent self-cleaning performance, which is of great significance for the control of marine biological pollution. As shown in Figure 6e and f, the D: K=0.7: 1.3/1.0 films show excellent extensibility and toughness which can be stretched to nearly 280% of their original length and lift a weight of 1 kg reaction kettle.

3.7. Antibacterial properties

The antibacterial efficiency of the D:K=0.7:1.3/1.0 coating was evaluated using *E. coli* and *S. aureus* as the typical

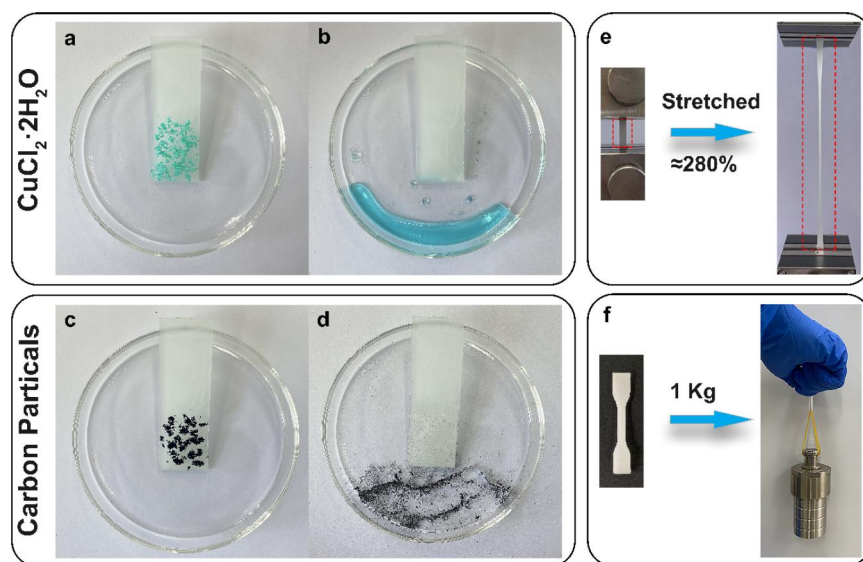


Figure 6. Self-cleaning test by $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$ particles: (a) before, (b) after. Self-cleaning test by carbon particles: (c) before, (d) after. (e) Photographs of a dumbbell-shaped specimen (D: K=0.7: 1.3/1.0) before and after stretching. (f) Photographs showing the ability to lift a 1 kg reaction kettle (mass of the specimen: 0.536 g).

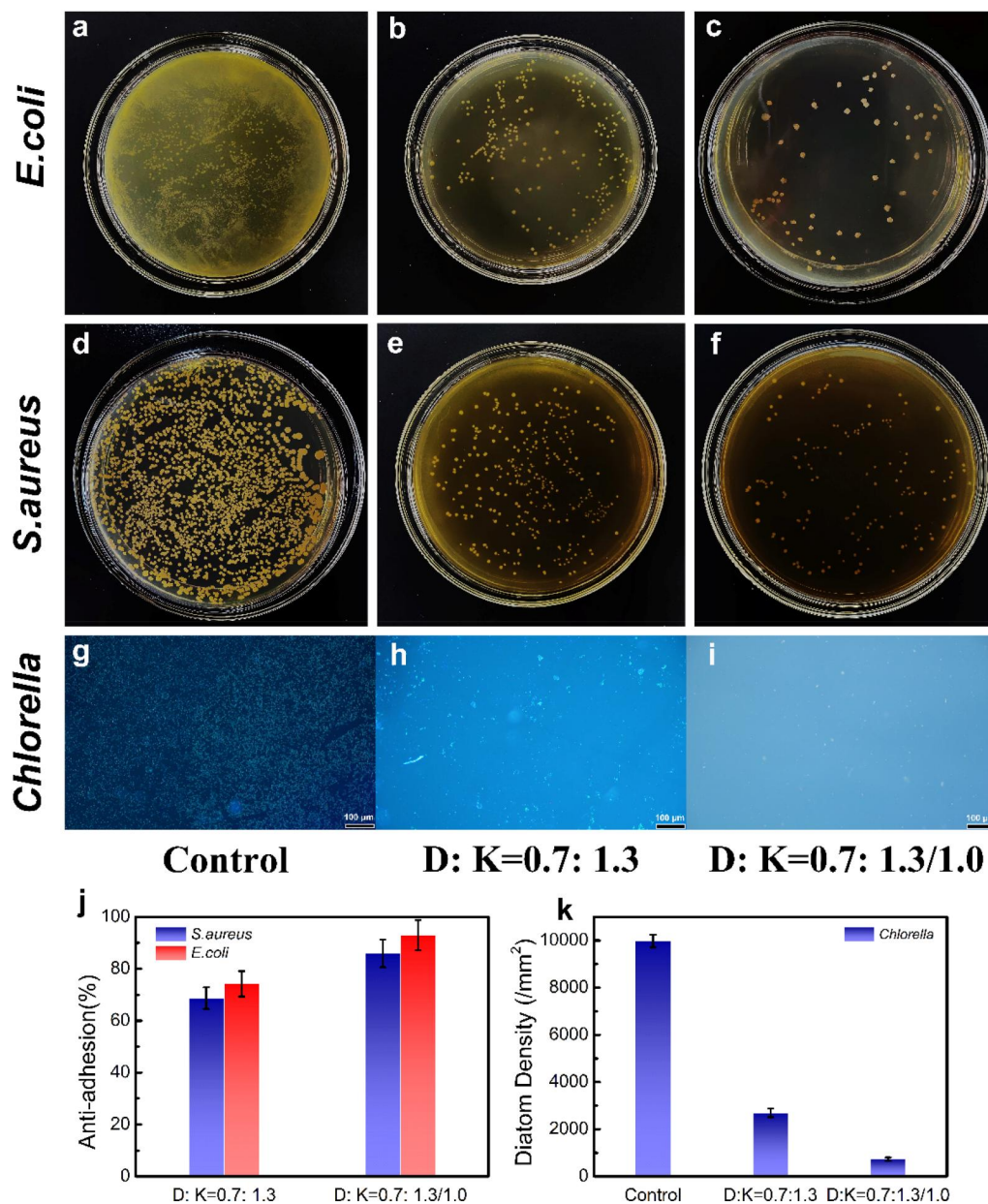


Figure 7. The marine antifouling properties of D: K=0.7: 1.3/1.0. Bacterial biofouling resistance test with *E. coli* after 24 h: (a) Control, (b) D: K=0.7: 1.3, (c) D: K=0.7: 1.3/1.0. Bacterial biofouling resistance test with *S. aureus* after 24 h: (d) Control, (e) D: K=0.7: 1.3, (f) D: K=0.7: 1.3/1.0. Fluorescence images of *Chlorella* adhered on (g) Control, (h) D: K=0.7: 1.3, (i) D: K=0.7: 1.3/1.0. (j) Quantitative evaluation of anti-adhesion rates of *E. coli* and *S. aureus*. The corrosion resistance of D: K=0.7: 1.3/1.0: (h) before, (i) after 24 h.

fouling organisms. Adhesive bacteria on blank slides surface were used as a control sample. As shown in Figure 7a–f, compared with the coating of control and D: K=0.7: 1.3, the surface coated with D: K=0.7: 1.3/1.0 coating showed a significant decrease in bacterial count, which is due to the strong antibacterial effect of TCS. The antibacterial efficiency of D: K=0.7: 1.3 and D: K=0.7: 1.3/1.0 were calculated according to the formula (2). As shown in Figure 7j, the antibacterial efficiency of D: K=0.7: 1.3 against *E. coli* and *S. aureus* reached 92.9%±5.8% and 85.9%±5.3%, respectively. The anti-adhesion rates for *E. coli* and *S. aureus* were 74.2%±4.9% and 68.7%±4.2%, respectively.

3.8. Diatom settlement

To evaluate the anti-algae properties of the coatings, the anti-algae tests were conducted using *Chlorella*. Figure 7g and k clearly show that a large number of *Chlorella* adhere on the surface of glass (9978±268 diatoms mm⁻²). Figure 7h and k illustrate the number of *Chlorella* adhered on D: K=0.7: 1.3 (2700±186 diatoms mm⁻²) is lower than that on the hydrophilic glass surface, which may be due to the low surface energy of the coating. With the introduction of TCS content, the number of *Chlorella* attached to the coating decreases, reaching an extremely low value of 727±76 diatom mm⁻² on D: K=0.7: 1.3/1.0 (Figure 7i and k).

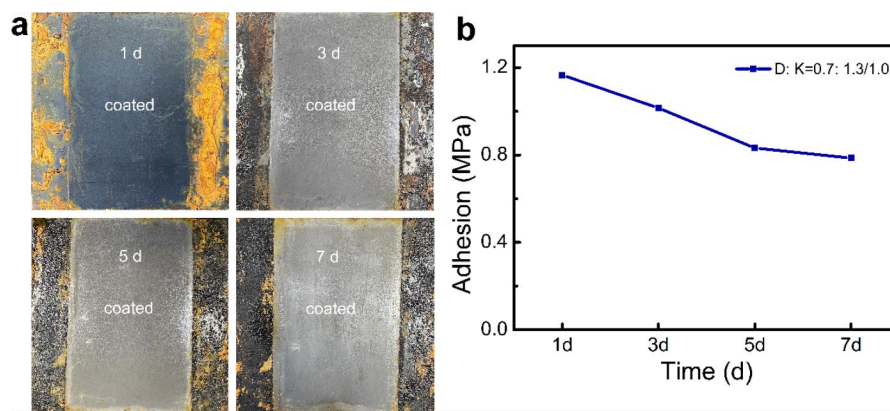


Figure 8. The corrosion resistance of D: K = 0.7: 1.3/1.0. (a) degree of corrosion of steel coated with D: K = 0.7: 1.3/1.0 for 1 d, 3 d, 5 d, and 7 d, (b) adhesion of D: K = 0.7: 1.3/1.0 for 1 d, 3 d, 5 d, and 7 d.

3.9. Corrosion resistance

As shown in Figure 8a, the actual corrosion resistance of D: K = 0.7: 1.3/1.0 was verified through salt spray corrosion tests. The D: K = 0.7: 1.3/1.0 is coated on the middle of the steel and the two sides of the steel are bare. 3.5 wt% NaCl artificial seawater was filled in the machine which mimics the conditions in the real marine environment. In Figure 8a, after 1 d, 3 d, 5 d, and 7 d, the coated part showed no signs of corrosion, but the coated parts on both sides were severely corroded, with a large number of corrosion products on the surface. In addition, the coated part does not see the coating peel off, which also proves that the adhesion of the coating meets the actual application environment even in harsh environments. Furthermore, the adhesion of the coatings to steel after 1 d, 3 d, 5 d, and 7 d are shown in Figure 8b, with the extension of the test time, the adhesion between the coating and steel gradually decreased, but the decrease was not too large. These can prove that the coating possesses a good performance of corrosion resistance.

4. Conclusions

In this study, we employed a simple method to synthesize a novel silicone-based polyurethane composite with TCS. Under static conditions, this polymer composite material exhibits excellent ultimate tensile strength, high fracture elongation efficiency, and good resistance to biological contamination. The chain extender DMBA prolongs the length of the molecular chains, enhancing the ultimate tensile strength. TCS provides the coating with antimicrobial properties. In conclusion, a successful development of silicone-based polyurethane antifouling coatings with improved adhesion, high mechanical strength, and excellent static anti-fouling performance has been achieved, making them promising for application in marine anti biofouling.

Supporting information

The following files are available free of charge.

Figures of WCA, SE, adhesion strength, the stress and strain curves, shore Hardness and swelling ratio of D:K = 1:1, D:K = 0.7:1.3, D:K = 0.5:1.5, and D:K = 0.3:1.7.

Table of the formulation of samples, static contact angles and surface energy, and adhesion strength of samples.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported financially by the New Colleges and Universities Twenty Foundational Projects of Jinan City (grant No. 2021GXRC068), the basic research projects of science, education, and industry integration pilot engineering of Qilu University of Technology (Shandong Academy of Sciences) (grant No. 2023PY002), the talent research projects of Qilu University of Technology (Shandong Academy of Sciences) (grant No. 2023RCKY013), the Natural Science Foundation of Shandong Province (grant No. ZR2022MB143), the 2023 graduate education and teaching reform projects of Qilu University of Technology (Shandong Academy of Sciences) (grant No. 24231601010415), the major innovation projects for integrating science, education & industry of Qilu University of Technology (Shandong Academy of Sciences) (grant No. 2022JBZ01-07).

References

- [1] Jin, H.; Tian, L.; Bing, W.; Zhao, J.; Ren, L. Bioinspired Marine Antifouling Coatings: Status, Prospects, and Future. *Prog. Mater. Sci.* **2022**, *124*, 100889. DOI: [10.1016/j.pmatsci.2021.100889](https://doi.org/10.1016/j.pmatsci.2021.100889).
- [2] Li, M.-Y.; Zhang, B.-X.; Chen, L.-L.; Kong, J.-J.; Wang, Y.-F.; Ba, M. Spontaneous Directed Leaching of Slippery Lubricant in Self-Stratified Marine Antifouling Coatings for Nonporous Surfaces. *Prog. Org. Coat.* **2023**, *184*, 107858. DOI: [10.1016/j.porgcoat.2023.107858](https://doi.org/10.1016/j.porgcoat.2023.107858).
- [3] Wang, M.; Ma, C. F.; Zhang, G. Z. Dynamic Surface Antifouling Materials. *Chin. J. Chem.* **2023**, *41*, 2881–2888. DOI: [10.1002/cjoc.202300321](https://doi.org/10.1002/cjoc.202300321).
- [4] Weber, F.; Esmaili, N. Marine Biofouling and the Role of Biocidal Coatings in Balancing Environmental Impacts. *Biofouling* **2023**, *39*, 661–681. DOI: [10.1080/08927014.2023.2246906](https://doi.org/10.1080/08927014.2023.2246906).
- [5] Xie, Q.; Pan, J.; Ma, C.; Zhang, G. Dynamic Surface Antifouling: Mechanism and Systems. *Soft Matter* **2019**, *15*, 1087–1107. DOI: [10.1039/c8sm01853g](https://doi.org/10.1039/c8sm01853g).
- [6] Zeng, Y.; Wang, P.; He, B.; Liu, S.; Ye, Q.; Zhou, F. Fabrication of Zwitterionic Polymer-Functionalized MXene Nanosheets for Anti-Bacterial and anti-Biofouling Applications. *Prog. Org. Coat.* **2023**, *183*, 107727. DOI: [10.1016/j.porgcoat.2023.107727](https://doi.org/10.1016/j.porgcoat.2023.107727).

- [7] Wang, A.; De Silva, K.; Jones, M.; Robinson, P.; Larribe, G.; Gao, W. Anticorrosive Coating Systems for Marine Propellers. *Prog. Org. Coat.* **2023**, *183*, 107768. DOI: [10.1016/j.porgcoat.2023.107768](https://doi.org/10.1016/j.porgcoat.2023.107768).
- [8] Blackwood, D. J.; Lim, C. S.; Teo, S. L. M.; Hu, X.; Pang, J. Macrofouling Induced Localized Corrosion of Stainless Steel in Singapore Seawater. *Corros. Sci.* **2017**, *129*, 152–160. DOI: [10.1016/j.corsci.2017.10.008](https://doi.org/10.1016/j.corsci.2017.10.008).
- [9] Fitridge, I.; Dempster, T.; Guenther, J.; de Nys, R. The Impact and Control of Biofouling in Marine Aquaculture: A Review. *Biofouling* **2012**, *28*, 649–669. DOI: [10.1080/08927014.2012.700478](https://doi.org/10.1080/08927014.2012.700478).
- [10] Li, H.; Cui, Z.; Cui, H.; Bai, Y.; Yin, Z.; Qu, K. Hazardous Substances and Their Removal in Recirculating Aquaculture Systems: A Review. *Aquaculture* **2023**, *569*, 739399. DOI: [10.1016/j.aquaculture.2023.739399](https://doi.org/10.1016/j.aquaculture.2023.739399).
- [11] Bannister, J.; Sievers, M.; Bush, F.; Bloecher, N. Biofouling in Marine Aquaculture: A Review of Recent Research and Developments. *Biofouling* **2019**, *35*, 631–648. DOI: [10.1080/08927014.2019.1640214](https://doi.org/10.1080/08927014.2019.1640214).
- [12] Eduok, U.; Faye, O.; Szpunar, J. Recent Developments and Applications of Protective Silicone Coatings: A Review of PDMS Functional Materials. *Prog. Org. Coat.* **2017**, *111*, 124–163. DOI: [10.1016/j.porgcoat.2017.05.012](https://doi.org/10.1016/j.porgcoat.2017.05.012).
- [13] Hu, P.; Xie, Q.; Ma, C.; Zhang, G. Silicone-Based Fouling-Release Coatings for Marine Antifouling. *Langmuir* **2020**, *36*, 2170–2183. DOI: [10.1021/acs.langmuir.9b03926](https://doi.org/10.1021/acs.langmuir.9b03926).
- [14] Guo, H.; Song, L.; Hu, J.; Lin, T.; Li, X.; Yu, H.; Cheng, D.; Hou, Y.; Zhan, X.; Zhang, Q.; et al. Enhanced Antifouling Strategy with a Strong Synergistic Effect of Fluorescent Antifouling and Contact Bacteriostasis Using 7-Amino-4-Methylcoumarin. *Chem. Eng. J.* **2021**, *420*, 127676. DOI: [10.1016/j.cej.2020.127676](https://doi.org/10.1016/j.cej.2020.127676).
- [15] Amara, I.; Miled, W.; Slama, R. B.; Ladhari, N. Antifouling Processes and Toxicity Effects of Antifouling Paints on Marine Environment. A Review. *Environ. Toxicol. Pharmacol.* **2018**, *57*, 115–130. DOI: [10.1016/j.etap.2017.12.001](https://doi.org/10.1016/j.etap.2017.12.001).
- [16] Lin, X.; Xie, Q.; Ma, C.; Zhang, G. Self-Healing, Highly Elastic and Amphiphilic Silicone-Based Polyurethane for Antifouling Coatings. *J. Mater. Chem. B* **2021**, *9*, 1384–1394. DOI: [10.1039/D0TB02465A](https://doi.org/10.1039/D0TB02465A).
- [17] Dafforn, K. A.; Lewis, J. A.; Johnston, E. L. Antifouling Strategies: History and Regulation, Ecological Impacts and Mitigation. *Mar. Pollut. Bull.* **2011**, *62*, 453–465. DOI: [10.1016/j.marpolbul.2011.01.012](https://doi.org/10.1016/j.marpolbul.2011.01.012).
- [18] Gittens, J. E.; Smith, T. J.; Suleiman, R.; Akid, R. Current and Emerging Environmentally-Friendly Systems for Fouling Control in the Marine Environment. *Biotechnol. Adv.* **2013**, *31*, 1738–1753. DOI: [10.1016/j.biotechadv.2013.09.002](https://doi.org/10.1016/j.biotechadv.2013.09.002).
- [19] Pistone, A.; Scolaro, C.; Visco, A. Mechanical Properties of Protective Coatings against Marine Fouling: A Review. *Polymers (Basel)* **2021**, *13*, 173. DOI: [10.3390/polym13020173](https://doi.org/10.3390/polym13020173).
- [20] Hu, P.; Xie, Q.; Ma, C.; Zhang, G. Fouling Resistant Silicone Coating with Self-Healing Induced by Metal Coordination. *Chem. Eng. J.* **2021**, *406*, 126870. DOI: [10.1016/j.cej.2020.126870](https://doi.org/10.1016/j.cej.2020.126870).
- [21] Tribou, M.; Swain, G. The Use of Proactive in-Water Grooming to Improve the Performance of Ship Hull Antifouling Coatings. *Biofouling* **2010**, *26*, 47–56. DOI: [10.1080/08927010903290973](https://doi.org/10.1080/08927010903290973).
- [22] Lejars, M.; Margailan, A.; Bressy, C. Fouling Release Coatings: A Nontoxic Alternative to Biocidal Antifouling Coatings. *Chem. Rev.* **2012**, *112*, 4347–4390. DOI: [10.1021/cr200350v](https://doi.org/10.1021/cr200350v).
- [23] Galhenage, T. P.; Webster, D. C.; Moreira, A. M. S.; Burgett, R. J.; Stafslie, S. J.; Vanderwal, L.; Finlay, J. A.; Franco, S. C.; Clare, A. S. Poly(Ethylene) Glycol-Modified, Amphiphilic, Siloxane-Polyurethane Coatings and Their Performance as Fouling-Release Surfaces. *J. Coat. Technol. Res.* **2017**, *14*, 307–322. DOI: [10.1007/s11998-016-9862-9](https://doi.org/10.1007/s11998-016-9862-9).
- [24] Molino, P. J.; Childs, S.; Eason Hubbard, M. R.; Carey, J. M.; Burgman, M. A.; Wetherbee, R. Development of the Primary Bacterial Microfouling Layer on Antifouling and Fouling Release Coatings in Temperate and Tropical Environments in Eastern Australia. *Biofouling* **2009**, *25*, 149–162. DOI: [10.1080/08927010802592917](https://doi.org/10.1080/08927010802592917).
- [25] Holland, R.; Dugdale, T. M.; Wetherbee, R.; Brennan, A. B.; Finlay, J. A.; Callow, J. A.; Callow, M. E. Adhesion and Motility of Fouling Diatoms on a Silicone Elastomer. *Biofouling* **2004**, *20*, 323–329. DOI: [10.1080/08927010400029031](https://doi.org/10.1080/08927010400029031).
- [26] Molino, P. J.; Campbell, E.; Wetherbee, R. Development of the Initial Diatom Microfouling Layer on Antifouling and Fouling-Release Surfaces in Temperate and Tropical Australia. *Biofouling* **2009**, *25*, 685–694. DOI: [10.1080/08927010903089912](https://doi.org/10.1080/08927010903089912).
- [27] Kausar, A. Polyurethane/Polyhedral Oligomeric Silsesquioxane Nanocomposite: Trends and Perspectives. *J. Macromol. Sci. Part A-Pure Appl. Chem.* **2021**, *58*, 361–375. DOI: [10.1080/10601325.2020.1866437](https://doi.org/10.1080/10601325.2020.1866437).
- [28] Chen, M.; Ou, B.; Guo, Y.; Guo, Y.; Kang, Y.; Liu, H.; Yan, J.; Tian, L. Preparation of an Environmentally Friendly Antifouling Degradable Polyurethane Coating Material Based on Medium-Length Fluorinated Diols. *J. Macromol. Sci. Part A-Pure Appl. Chem.* **2018**, *55*, 483–488. DOI: [10.1080/10601325.2018.1470466](https://doi.org/10.1080/10601325.2018.1470466).
- [29] Hu, P.; Xie, R.; Xie, Q.; Ma, C.; Zhang, G. Simultaneous Realization of Antifouling, Self-Healing, and Strong Substrate Adhesion via a Bioinspired Self-Stratification Strategy. *Chem. Eng. J.* **2022**, *449*, 137875. DOI: [10.1016/j.cej.2022.137875](https://doi.org/10.1016/j.cej.2022.137875).
- [30] Sun, J.; Liu, C.; Duan, J.; Liu, J.; Dong, X.; Zhang, Y.; Wang, N.; Wang, J.; Hou, B. Facile Fabrication of Self-Healing Silicone-Based Poly(Urea-Thiourea)/Tannic Acid Composite for anti-Biofouling. *J. Mater. Sci. Technol.* **2022**, *124*, 1–13. DOI: [10.1016/j.jmst.2022.01.026](https://doi.org/10.1016/j.jmst.2022.01.026).
- [31] Zhao, Z.; Liu, Q.; Liao, Y.; Yu, P.; Tang, Y.; Liu, Q.; Shi, X.; Shou, L.; Zeng, J.; Chen, Q.; et al. Ecological Risk Assessment of Trace Metals in Sediments and Their Effect on Benthic Organisms from the South Coast of Zhejiang Province, China. *Mar. Pollut. Bull.* **2023**, *187*, 114529. DOI: [10.1016/j.marpolbul.2022.114529](https://doi.org/10.1016/j.marpolbul.2022.114529).
- [32] Poger, D.; Mark, A. E. Effect of Triclosan and Chloroxenol on Bacterial Membranes. *J. Phys. Chem. B* **2019**, *123*, 5291–5301. DOI: [10.1021/acs.jpcc.9b02588](https://doi.org/10.1021/acs.jpcc.9b02588).
- [33] Owens, D. K.; Wendt, R. C. Estimation of the Surface Free Energy of Polymers. *J. Appl. Polym. Sci.* **1969**, *13*, 1741–1747. (accessed 2023/04/26). DOI: [10.1002/app.1969.070130815](https://doi.org/10.1002/app.1969.070130815).
- [34] Wang, Y.; Hao, D.; Yang, M.; Su, X.; Li, P.; Liu, Q.; Guo, X. Polyurethane Antifouling Coatings with Various Antifouling Strategies in the Side Chain. *Prog. Org. Coat.* **2022**, *173*, 107225. DOI: [10.1016/j.porgcoat.2022.107225](https://doi.org/10.1016/j.porgcoat.2022.107225).
- [35] Zhang, K.; Song, C.; Wang, Z.; Gao, C.; Wu, Y.; Liu, Y. A Stretchable and Self-Healable Organosilicon Conductive Nanocomposite for a Reliable and Sensitive Strain Sensor. *J. Mater. Chem. C* **2020**, *8*, 17277–17288. DOI: [10.1039/D0TC04719H](https://doi.org/10.1039/D0TC04719H).
- [36] Zhang, K.; Gao, C.; Song, J.; Song, C.; Wang, Z.; Wu, Y.; Liu, Y.; Sun, J. A Type of Dissoluble Organosilicon Elastomer with Stretchable, Self-Healable and Antibacterial Properties. *Polym. Test.* **2021**, *96*, 107082. DOI: [10.1016/j.polymertesting.2021.107082](https://doi.org/10.1016/j.polymertesting.2021.107082).
- [37] Sun, J.; Duan, J.; Liu, X.; Dong, X.; Zhang, Y.; Liu, C.; Hou, B. Environmentally Benign Smart Self-Healing Silicone-Based Coating with Dual Antifouling and anti-Corrosion Properties. *Appl. Mater. Today* **2022**, *28*, 101551. DOI: [10.1016/j.apmt.2022.101551](https://doi.org/10.1016/j.apmt.2022.101551).
- [38] Gurunathan, T.; Chung, J. S. Physicochemical Properties of Amino-Silane-Terminated Vegetable Oil-Based Waterborne Polyurethane Nanocomposites. *ACS Sustainable Chem. Eng.* **2016**, *4*, 4645–4653. DOI: [10.1021/acssuschemeng.6b00768](https://doi.org/10.1021/acssuschemeng.6b00768).

- [39] Liu, Y.; Zhao, S.; Jiang, Y.; Liu, Q.; Liu, Z.; Song, C.; Wang, C. AgNP Hybrid Organosilicon Antifouling Coating with Spontaneous Self-Healing and Long-Term Antifouling Properties. *Ind. Eng. Chem. Res.* **2023**, *62*, 12999–13008. DOI: [10.1021/acs.iecr.3c01569](https://doi.org/10.1021/acs.iecr.3c01569).
- [40] Song, Y.; Qin, Q.; Jing, L.; Li, M.; Zhao, H.; Cui, J.; Zhang, Y. Design and Construction of Poly(Benzoquinone-Diamine)/Ti₃C₂TX Hybrid Electrode toward Advanced Aqueous Zinc-Ion Batteries. *Compos. Part B: Eng.* **2023**, *252*, 110517. DOI: [10.1016/j.compositesb.2023.110517](https://doi.org/10.1016/j.compositesb.2023.110517).
- [41] Brady, R. F. Properties Which Influence Marine Fouling Resistance in Polymers Containing Silicon and Fluorine. *Prog. Org. Coat.* **1999**, *35*, 31–35. DOI: [10.1016/S0300-9440\(99\)00005-3](https://doi.org/10.1016/S0300-9440(99)00005-3).
- [42] Baier, R. E. Surface Behaviour of Biomaterials: The Theta Surface for Biocompatibility. *J. Mater. Sci. Mater. Med.* **2006**, *17*, 1057–1062. DOI: [10.1007/s10856-006-0444-8](https://doi.org/10.1007/s10856-006-0444-8).
- [43] Kavanagh, C. J.; Quinn, R. D.; Swain, G. W. Observations of Barnacle Detachment from Silicones Using High-Speed Video. *J. Adhes.* **2005**, *81*, 843–868. DOI: [10.1080/00218460500189331](https://doi.org/10.1080/00218460500189331).